MODULE-4

Audio and video compression

AUDIO & VIDEO COMPRESSION

AUDIO COMPRESSION :

The digitization process of audio signals is known as pulse code Modulation or PCM. This involves sampling the audio signal / waveform at a minimum rate which is twice that of the max. freq. component that makes up the signal. If the BW of the commn channel to be used is less than that of the signal, then the sampling rate is determined by the BW of the commn channel. This is known as band limited signal.

Differential Pulse Code Modulation : (DPCM)

DPCM is a derivative of standard PCM & exploits the fact that, for most audio signals, the range of the differences in amplitude b/w successive samples of the audio waveform is less than the range of the actual sample amplitudes. Hence if only the digitized diff. signal is used to encode the waveform then fewer bits are required than for a comparable PCM signal. A DPCM encoder & decoder are shown in fig with simplified timing diagram.

The previous digitized sample of the analog i/p signal is held in the Register (R) & the diff. signal DPC[illegible] is computed by subtracting the current contents of the register (R_0) from the new digitized sample o/p by the ADC (PCM). The decoder operates by simply adding the

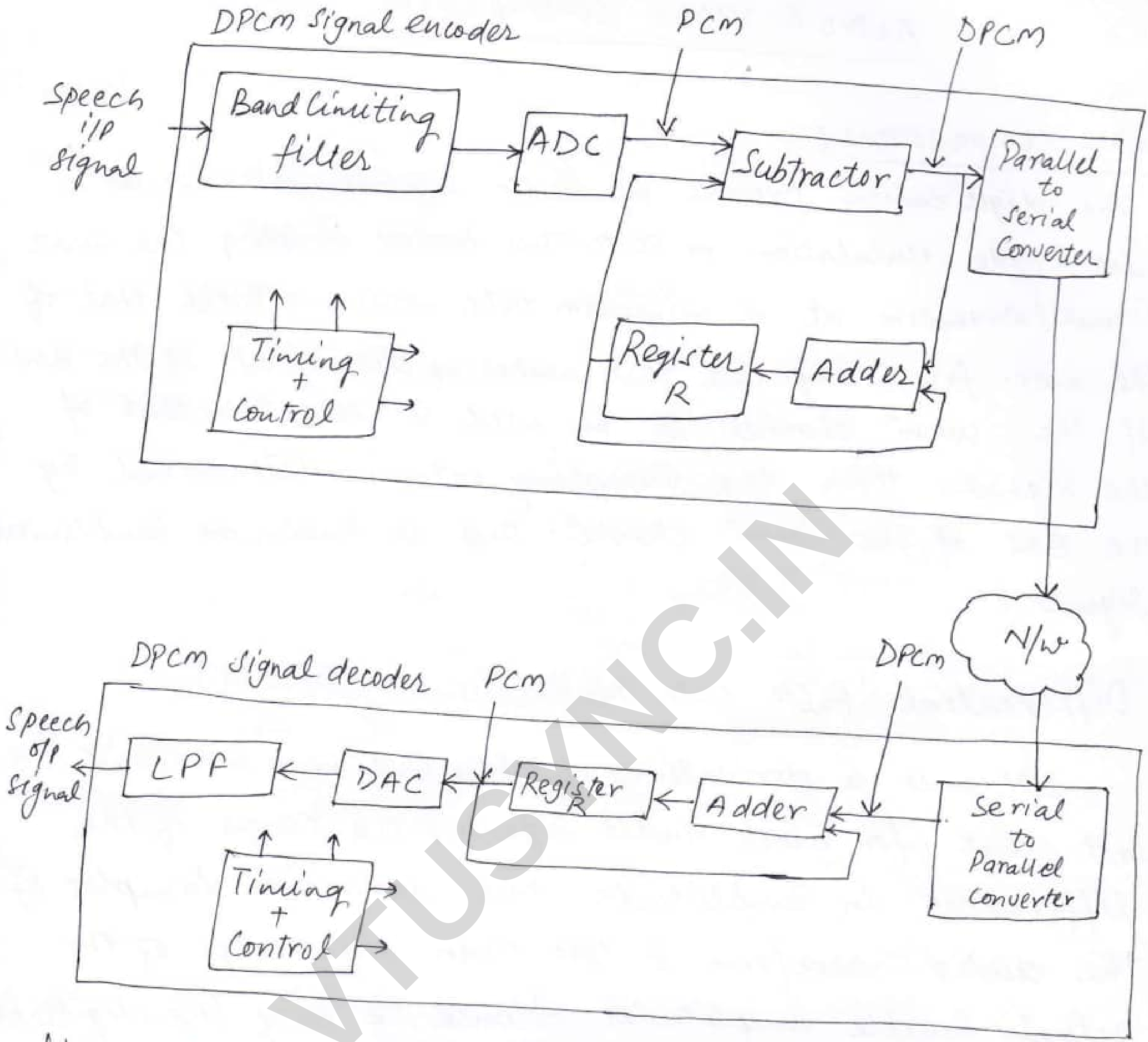
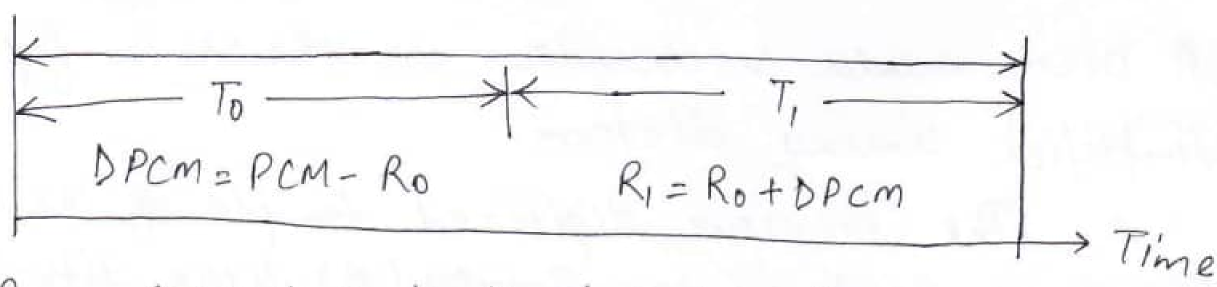


fig: DPCM: Encoder/Decoder

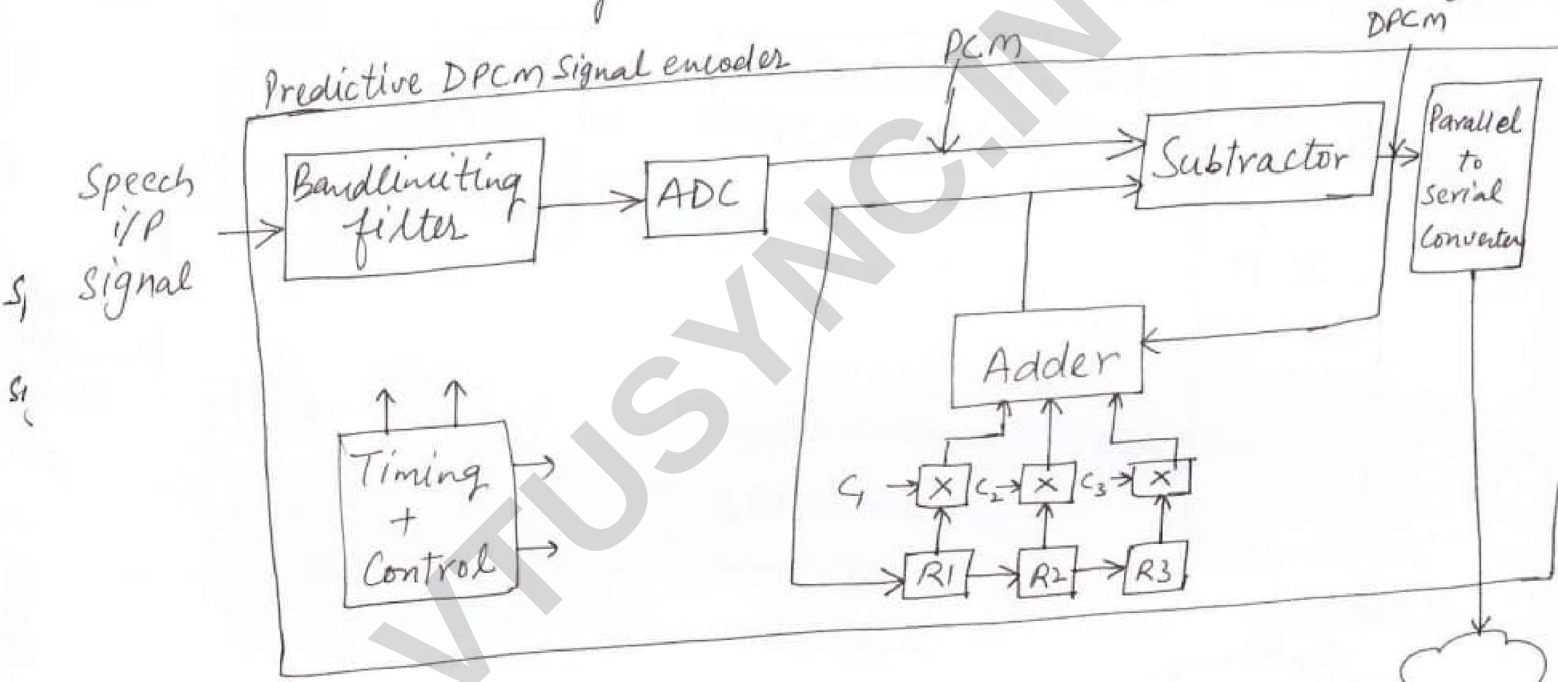


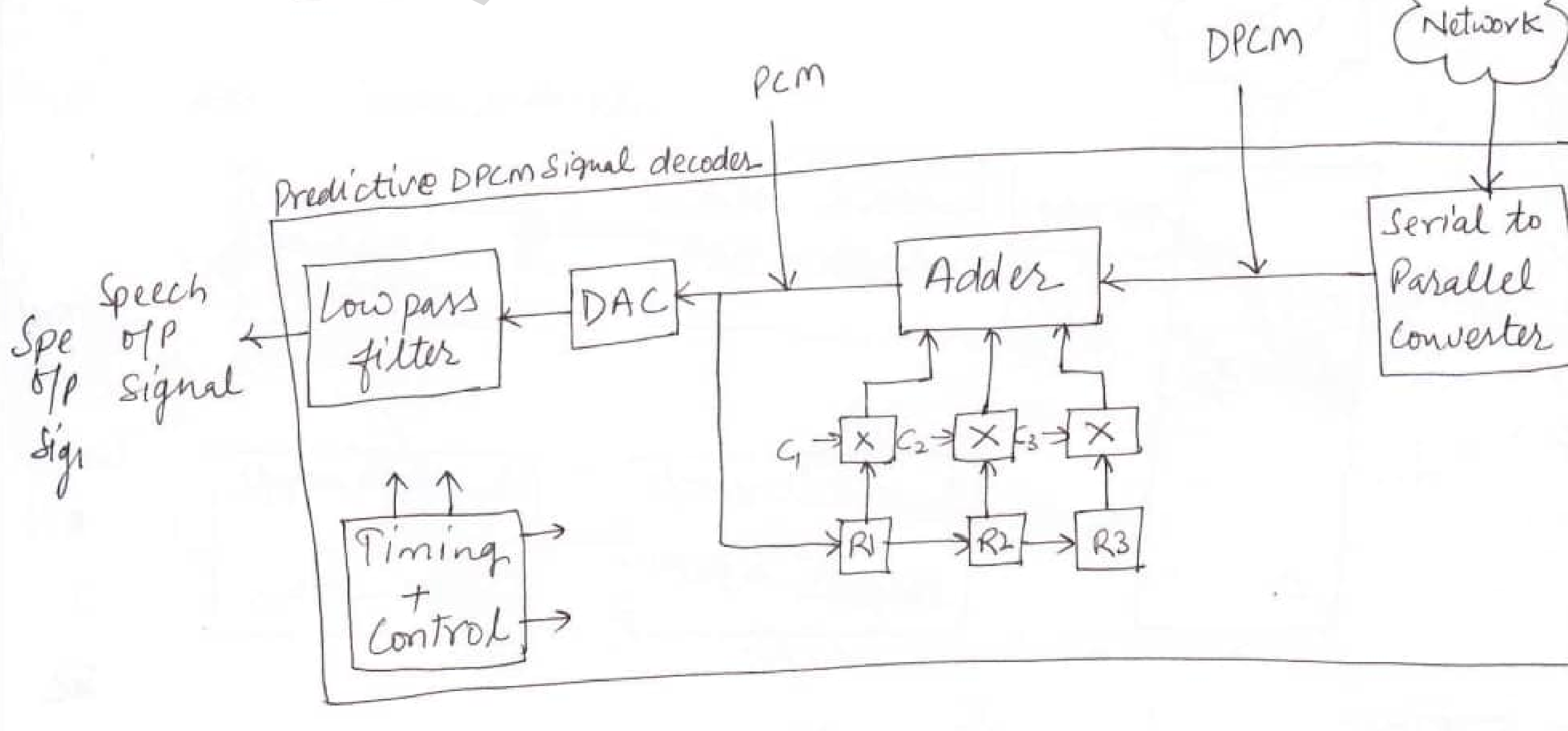
R_0 = Current contents of register R
R_1 = new/updated contents.

fig: encoder timing.

received difference signal (DPCM) to the previously computed signal held in the register (PCM). The output of the ADC is used directly and the accuracy of each computed diff. signal is known as residual signal. All ADC operations produce quantization errors.

With a basic DPCM ~~to~~ scheme, the previous value held in the register is only an approximation known as predicting. To achieve this prediction the previous signal by using not only the estimate of the current signal. The proportion used are determined by what are known as predictor coefficient

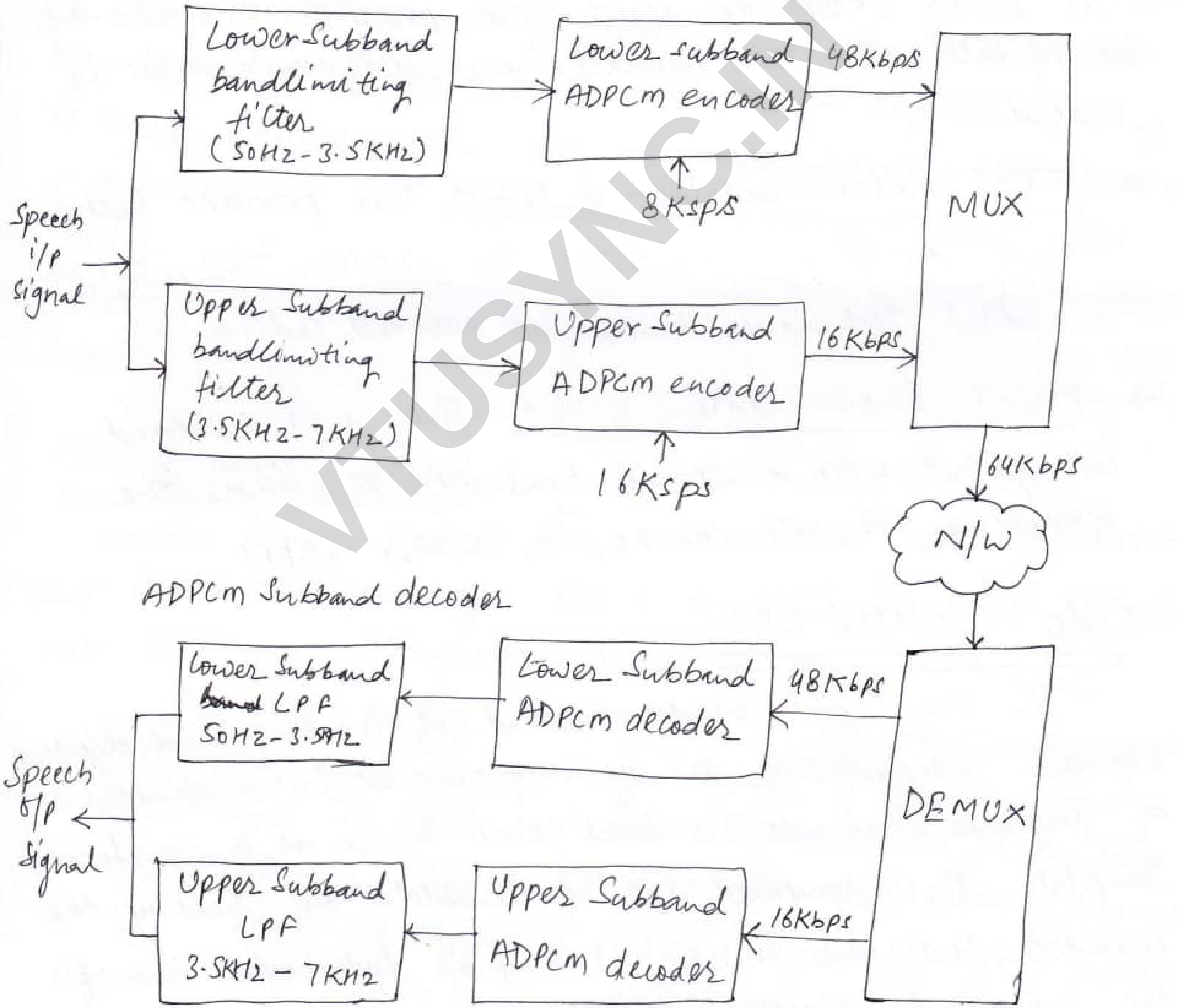




Adaptive differential PCM:

Saving in Bandwidth & improved quality can b obtained by varying the number of bits used for the difference signal depending on its amplitude i.e usin fewer bits to encode smaller difference values than for larger values. This is the principle of adaptive differen PCM. (ADPCM).

ADPCm Subband encoder



The input speech signal is effectively divided into two separate equal-BW signal, the first known as the lower subband signal & the 2nd the upper subband signal. Each is then sampled and encoded independently using ADPCM; The use of two subbands has the advantage that bit rates can be used for each.

There are three standard based on ADPCM

→ <u>ITU-T Recommendation G.721</u> :→ Uses the same principle as DPCM except an eight-order predictor is used & the no. of bits used to quantize each difference value is varied.

→ <u>ITU-T Recommendation G.722</u> :→ This provides better sound quality than G.721. To achieve this, it uses an added technique known as subband coding.

→ <u>ITU-T Recommendation G.726</u>: This uses subband coding but with a speech bandwidth of 3.4KHz. The operating bit rate can be 40, 32, 24 or 16Kbps.

Code - excited LPC:

In the CELP model, instead of treating each digitized segment independently for encoding purposes, just a limited set of ~~programs~~ segments is used, each known as a waveform template. A precomputed set of templetes are held by the encoder & decoder known as template Codebook. Each of the individual digitized samples that make up a particular template in the Codebook are differentially odeword that is sent selects a particular template from the codebook whose diff. values best match those quantized by the encoder. A continuity is maintained from one set of samples to another & an improvement in sound quality is obtained.

All coders of this type have a delay associated with them which is incurred while each block of digitized samples is analysed by the encoder & the speech is reconstructed at the decoder. The combined delay value is known as the coder's processing delay. The time to accumlate the block of samples is known as the algorithmic delay. To include samples from the next successive block, a technique is used known as lookahead. These delays occur in the coders and an end-to-end transmission delay over the network. The combined delay value of coder determines the suitability of coder for a specific application.

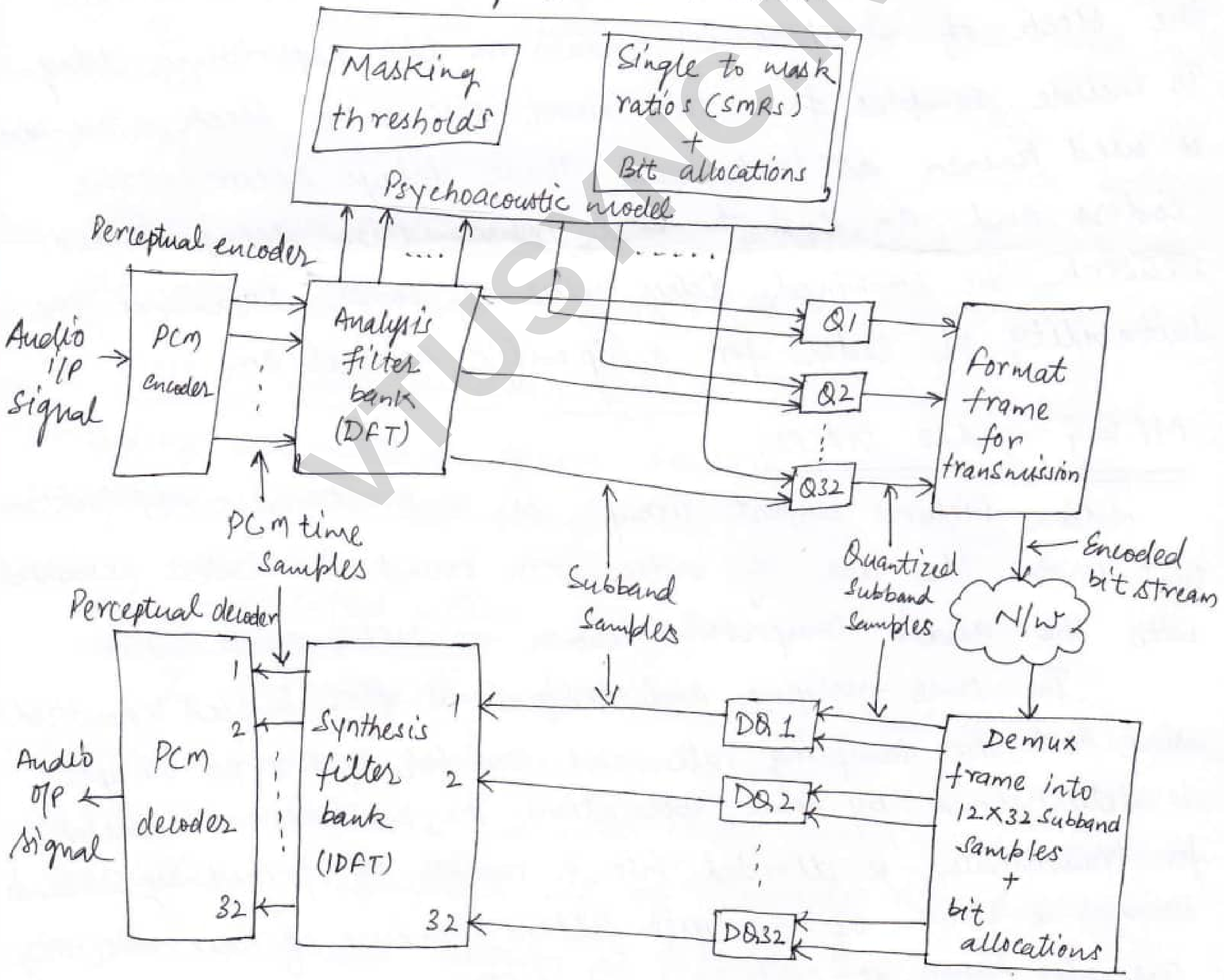
<u>MPEG audio coders:</u>

Motion Pictures Expert Group is a multimedia application that involve the use of video with sound. The coders associated with the audio compression known as MPEG audio coders.

The time-varying audio i/p signal is first sampled & quantized using PCM, the sampling rate and number of bits per sample is determined by some applications. The bandwidth i.e available for transmission is divided into a number of frequency subbands using a bank of analysis filters because of their role, they are also known as critical band filters.

Each frequency subband is of equal width & the bank of filters maps each set of 32 PCM samples into an equivalent set of 32 frequency samples, one per subband. Hence each is known as a Subband Sample.

In addition to filtering the input samples into separate freq. subbands, the analysis filter bank also determines the max. amplitude of 12 subband samples in each subband. Each is known as the scaling factor and these are passed both to the psycho-acoustic model & together with the set of freq. samples in each subband, to the corresponding quantizer block.

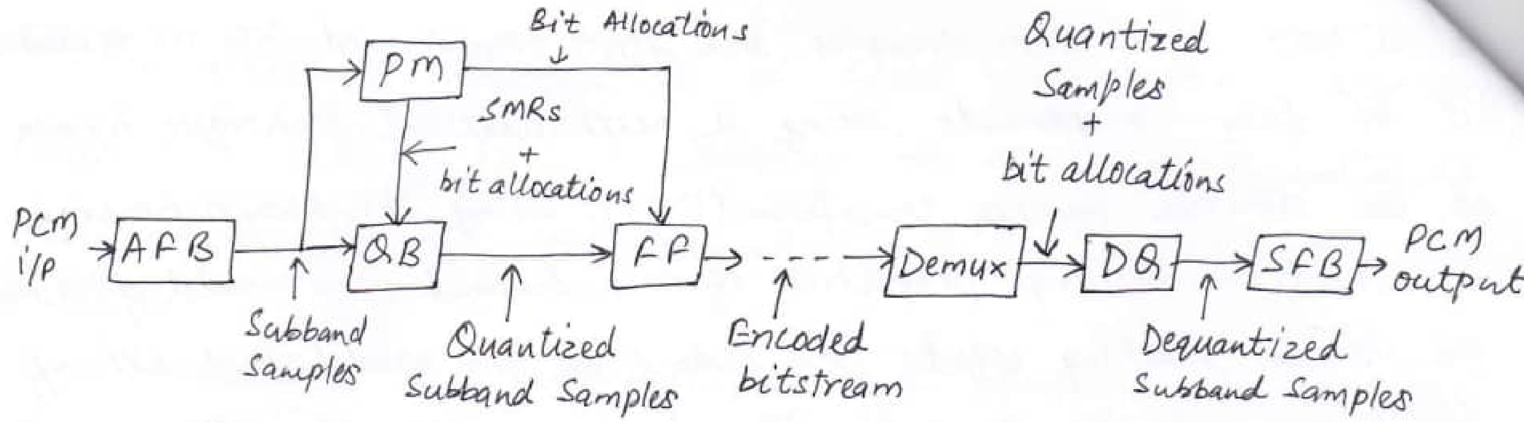


he 12 sets of 32 PCM samples are first transformed into an equivalent set of freq. components using a mathematical technique known as the discrete fourier transform (DFT). Using the known hearing thresholds & masking properties of each subband, the model determines the various masking effects. The output of the model is a set of signal to mask signals ratios (SMRs) and indicate those frequency components whose amplitude is below the related audible threshold.

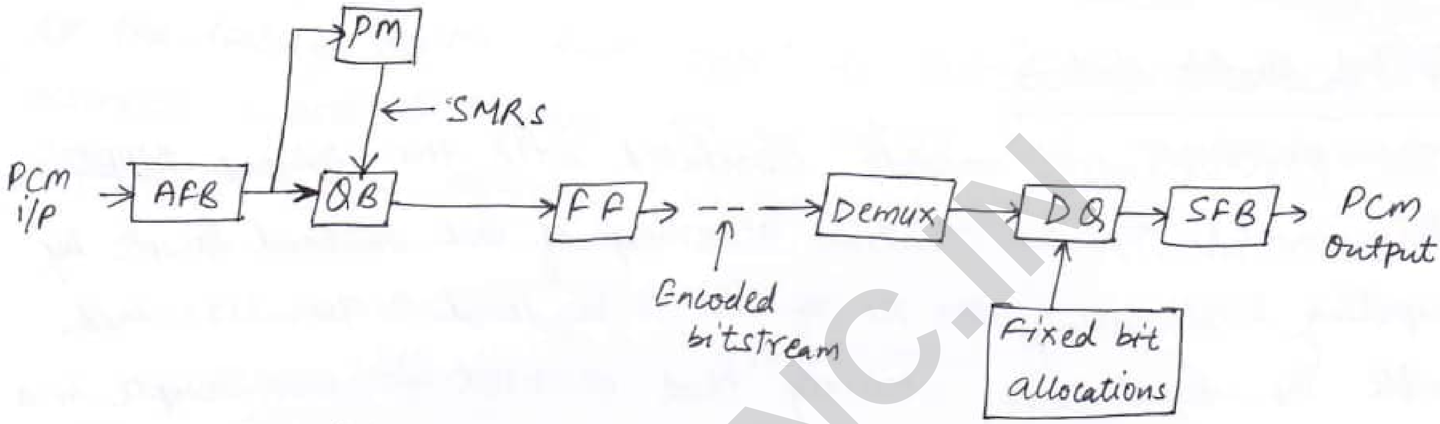
Dolby Audio Coders:

The psychoacoustic models associated with the various MPEG coders control the quantization accuracy of each subband sample by computing & allocating the no. of bits to be used to quantize each sample. The quantization accuracy that is used for each sample in a subband may very from one set of subband samples to the next, the bit allocation is used to quantize the samples in each subband is sent with the actual quantized samples. This information is then used by the decoder to dequantize the set of subband samples in the frame. This mode of operation of perceptual coder is known as the forward adaptive bit allocation mode. It has the disadvantage that a significant portion of each encoded frame contains bit allocation information leads to a inefficient use of available bit rate.

A variation of this approach is to use a fixed bit allocation strategy for each subband used by both encoder & decoder. The bit allocations that are selected for each subband determined by the sensitivity characteristics of the ear. This approach is used in a standard known as Dolby AC-1.



(a) Perceptual Coder: forward adaptive bit allocation (MPEG)



AFB = analysis filter bank QB = quantization blocks
SFB = Synthesis filter bank DQ = dequantization blocks
SMRs = Signal to mask ratios PM = Psychoacoustic model
FF = Frame formatter

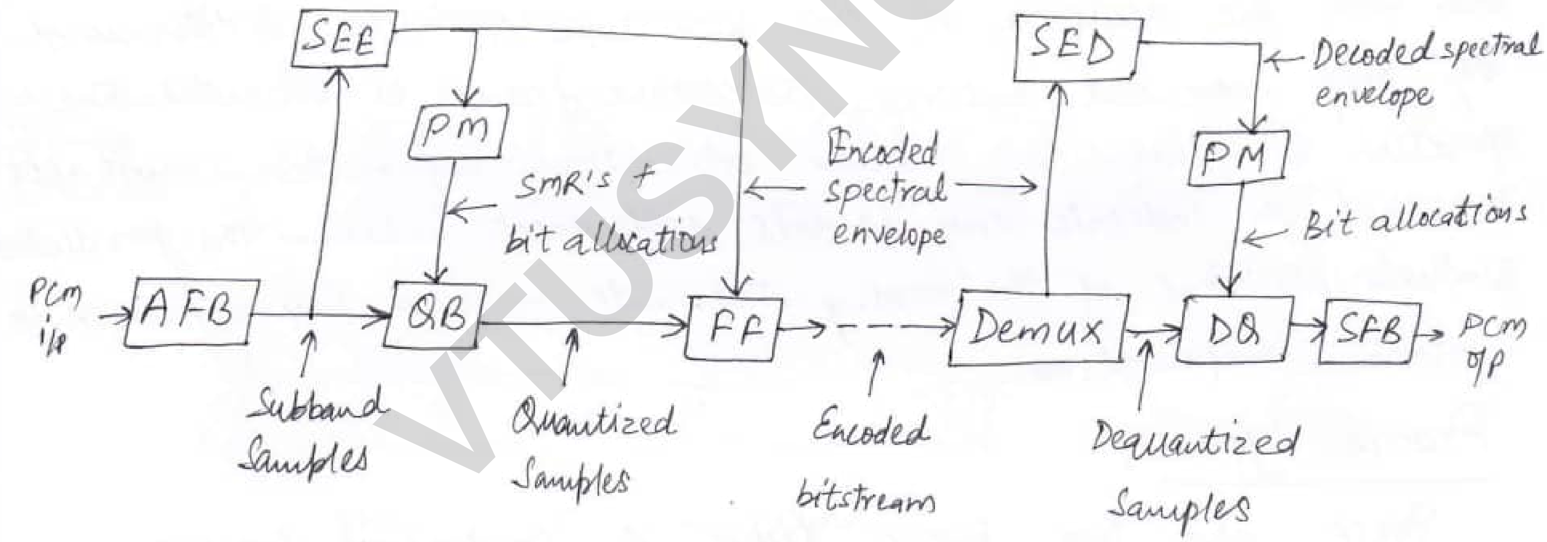
A second variation which allows the bit allocations per subband to be adaptive while at the same time minimizing the overheads in the encoder bitstream, for the decoder also to contain a copy of the psychoacoustic model. This is then used by the decoder to compute the same bit allocations that the psychoacoustic model in the encoder has used to quantize each set subband samples.

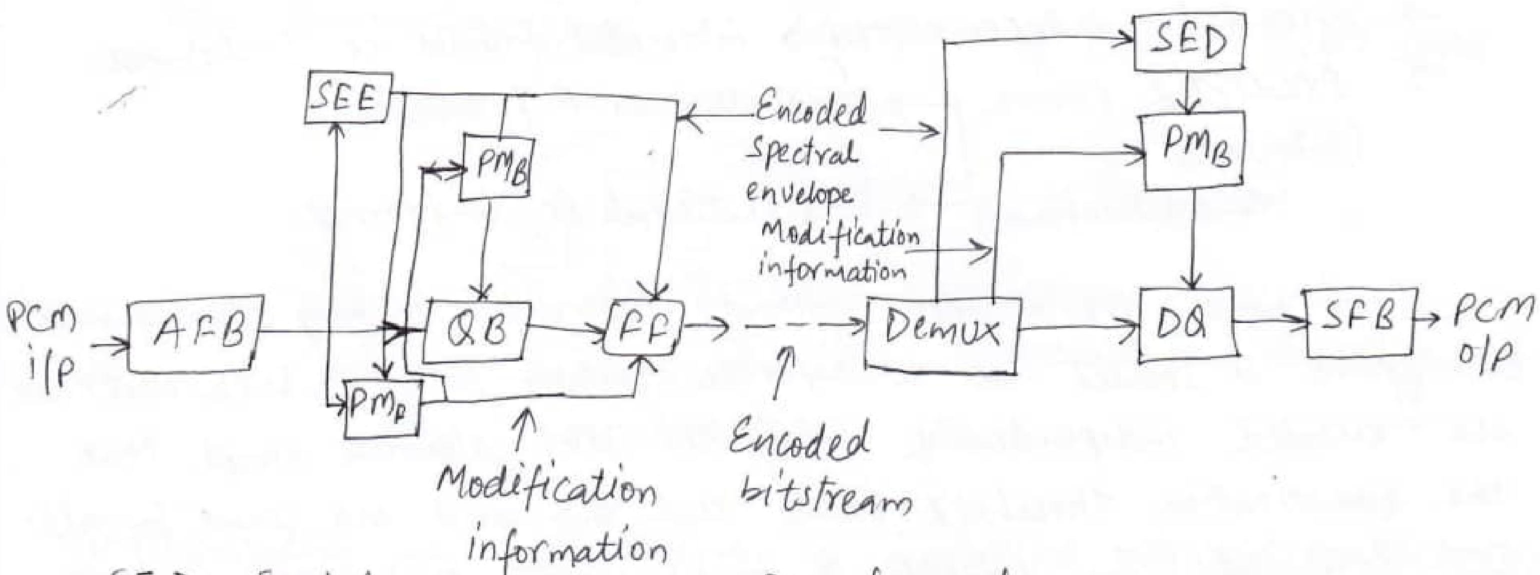
Hence with this operational mode, instead of each frame containing bit allocations information it contains the encoded frequency coefficients that are present in the sampled waveform segment. This is known as the encoded spectral envelope & this mode of operation the backward adaptive bit allocation mode illustrate the principle of operation of both the encoder & decoder.

Dolby AC-2 is utilized in many applications including the compression associated with the audio of a number of PC sound cards. They produce audio of hi-fi quality at a bit rate of 256 Kbps. For broadcast applications, it has the disadvantage that the same psychoacoustic model is required in decoder, the model in the encoder cannot be modified without changing all decoders.

A third variation has been developed that uses both backward/forward bit allocation principles. This is known as the hybrid backward/forward adaptive bit allocation mode.

AFB = analysis filter bank   SEE = Spectral envelope encoder
SFB = Synthesis filter bank   QB = quantization blocks





SED = Spectral envelope decoder   FF = frame formatter

VIDEO COMPRESSION:

Basic Principles:

Video is a sequence of digitized pictures ∴ video is referre[d] to as moving pictures. In principle, one approach to compressing a video source is to apply the JPEG algorithm to each frame independent[ly]. This approach is known as moving JPEG or MJPEG. The technique that is used to exploit the high correlation between successive frames i[s] to predict the content of many of the frames. This is based on a combination of a preceding frame. Instead of sending the source video as a set of individually-compressed frames, only the difference between the actual frame contents & the predicted frame contents are sent. The accuracy of the prediction operation is determined by any movement between successive frames is estimated. The operation is known as motion estimation. Information must also be sent to indicate any small differences between the predicte[d] & actual positions of the moving segments involved. This is known as motion compensation.

Frame types:

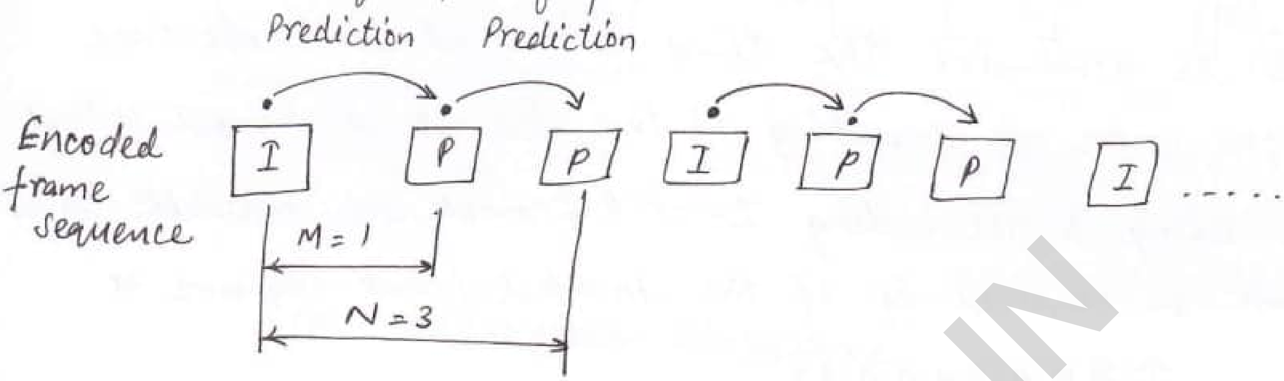
There are two basic types of Compressed frame:

→ encoded independently → intracoded frames or I-frames.

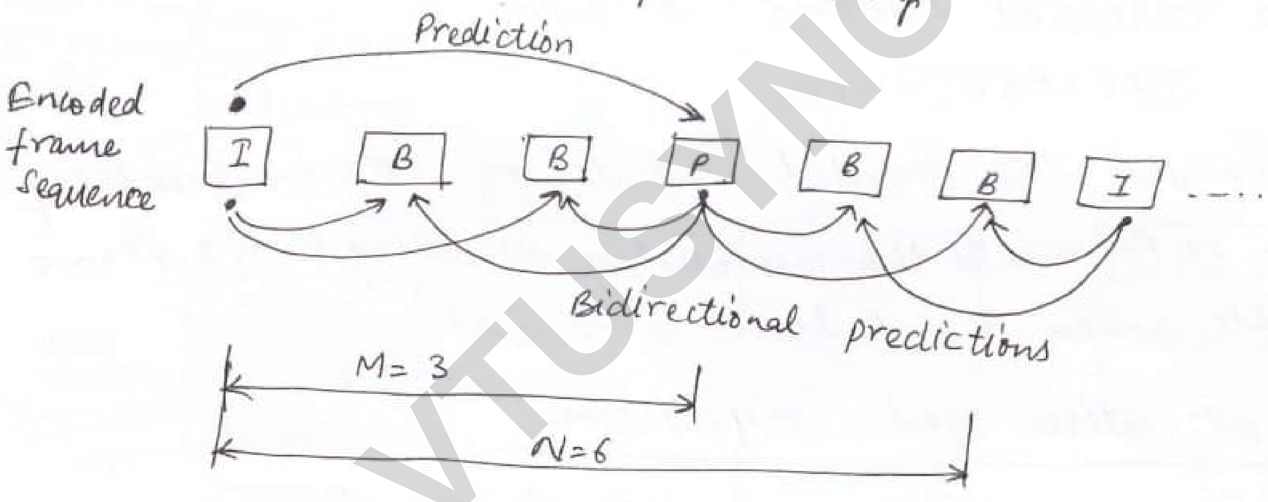
→ Predicted frame (intercoded or interpolation frames)
  - → predictive or P-frames
  - → Bidirectional or B-frames.

I-frames are encoded without reference to any other frames. Each frame is treated as a separate picture & Y, Cb & Cr matrices are encoded independently using the JPEG algorithm except that the quantization threshold values that are used are same for all DCT coefficients. In principle, it would appear to be best if these were limited to the first frame relating to a new scene in a movie

I-frames must be present in the output stream at regular intervals in order to allow for the possibility of the contents of an encoded I-frame being corrupted during transmission. I-frames are inserted into the output stream relatively frequently. The number of frames / pictures between successive I-frames is known as a group of pictures or GOP.

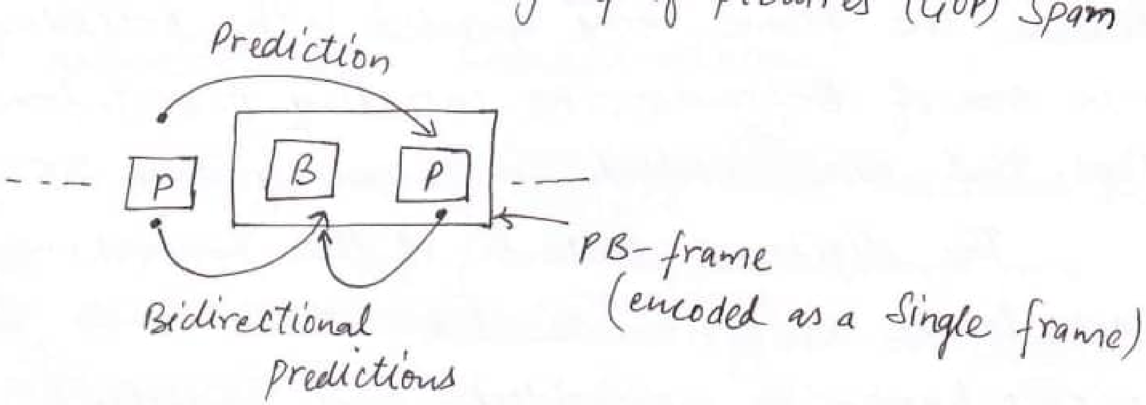


(a) I- & P- frames only



(b) I-, P- & B-frames

M = prediction Span   N = group of pictures (GOP) Spam



The encoding of a P-frame is relative to the contents of either a preceding I-frame or a preceding P-frame. P-frames are encoded using a combination of motion estimation & motion compensation. The no. of P-frames b/w each successive pair of I-frames is limited since any errors present in the first P-frame will be propagated to the next. The no. of frames b/w a P-frame & the immediately preceding I- or P-frame is called prediction span.

In order to minimize the time required to decode each B-frame, the order of encoding of the frames is changed so that both the preceding & succeeding I- or P-frames are available when the B-frame is received. Eg: if the uncoded frame sequence is

I B B P B B P B B I .....

then the reordered sequence would be:

I P B B P B B I B B P B B.

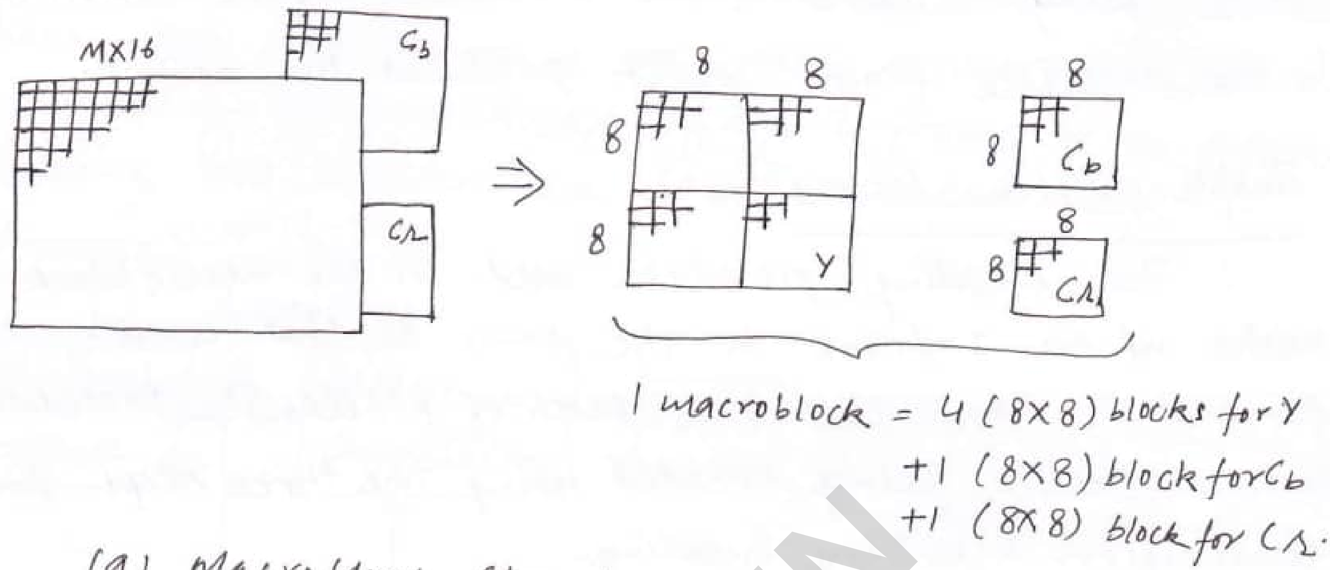
With B-frames, the decoded contents of both the immediately preceding I- or P-frame & the immediately succeding P- or I-frame are available when the B-frame is received.

Motion estimation and Compensation:

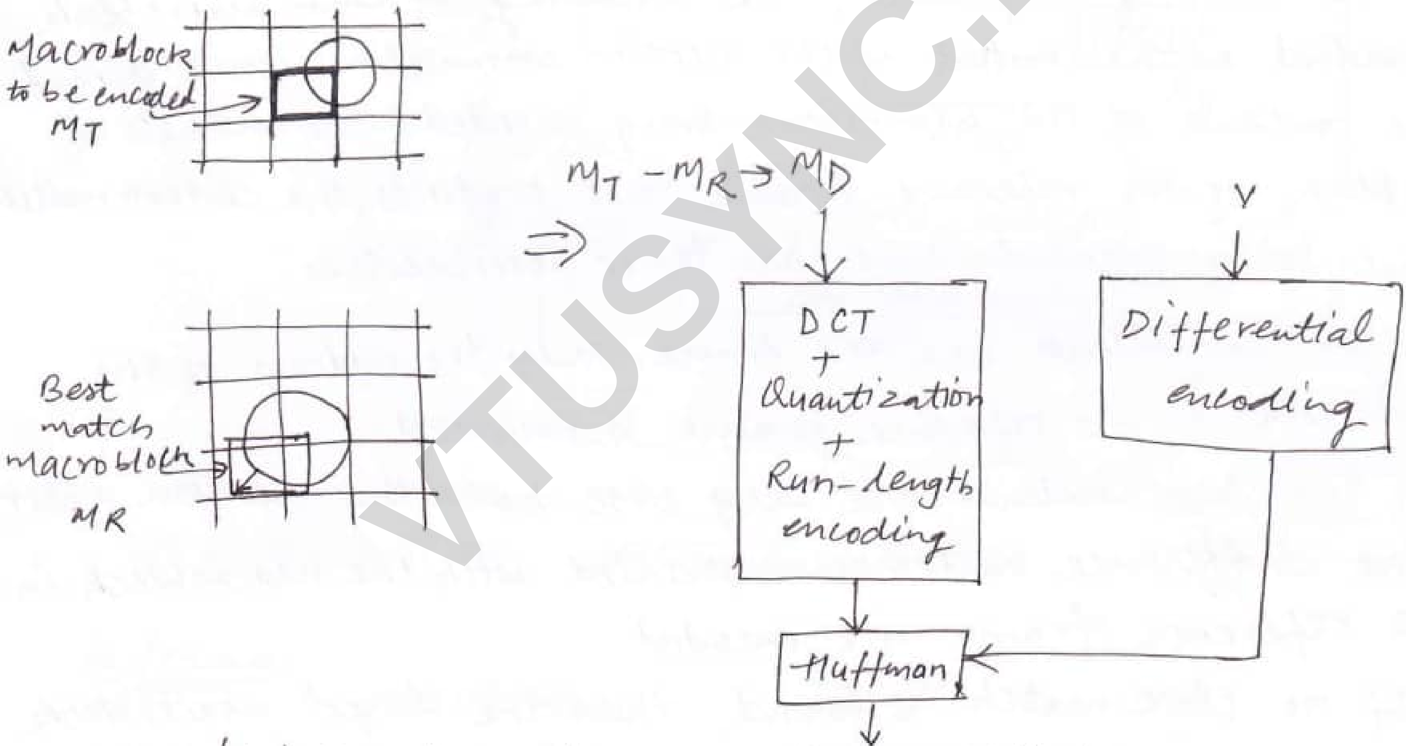
The encoded contents of both P- & B-frames are predicted by estimating any motion that has taken place between the frame being encoded & the preceding I- or P-frame & in case of B-frames, the succeding P- or I-frame. The various steps that are involved in encoding each P-frame.

The digitized contents of the Y-matrix associated with each frame are first divided into a 2-D matrix of 16 X 16 pixels known as macroblock. Each macroblock has an address associated with it & the block size used for the DCT operation is also 8 X 8 pixels, a macroblock comprises four DCT blocks for Luminance and one each for the two chrominance signals.

Video frame format (4:1:1)



(a) Macroblock Structure



(b) encoding Produre

To encode a P-frame, the contents of each maroblock in the frame known as the target frame, are compared on a pixel-by-pixel basis with the contents of the corresponding macroblock in the preceeding I or P-frame. This is known as reference frame.

All possible macroblock in the selected search area in the reference frame are searched for a match with a close match is found, two parameters are encoded. The first is known the motion vector & indicates the (x,y) offset of the macr block being encoded & the location of the block of pixels in the reference frame which produces the match.

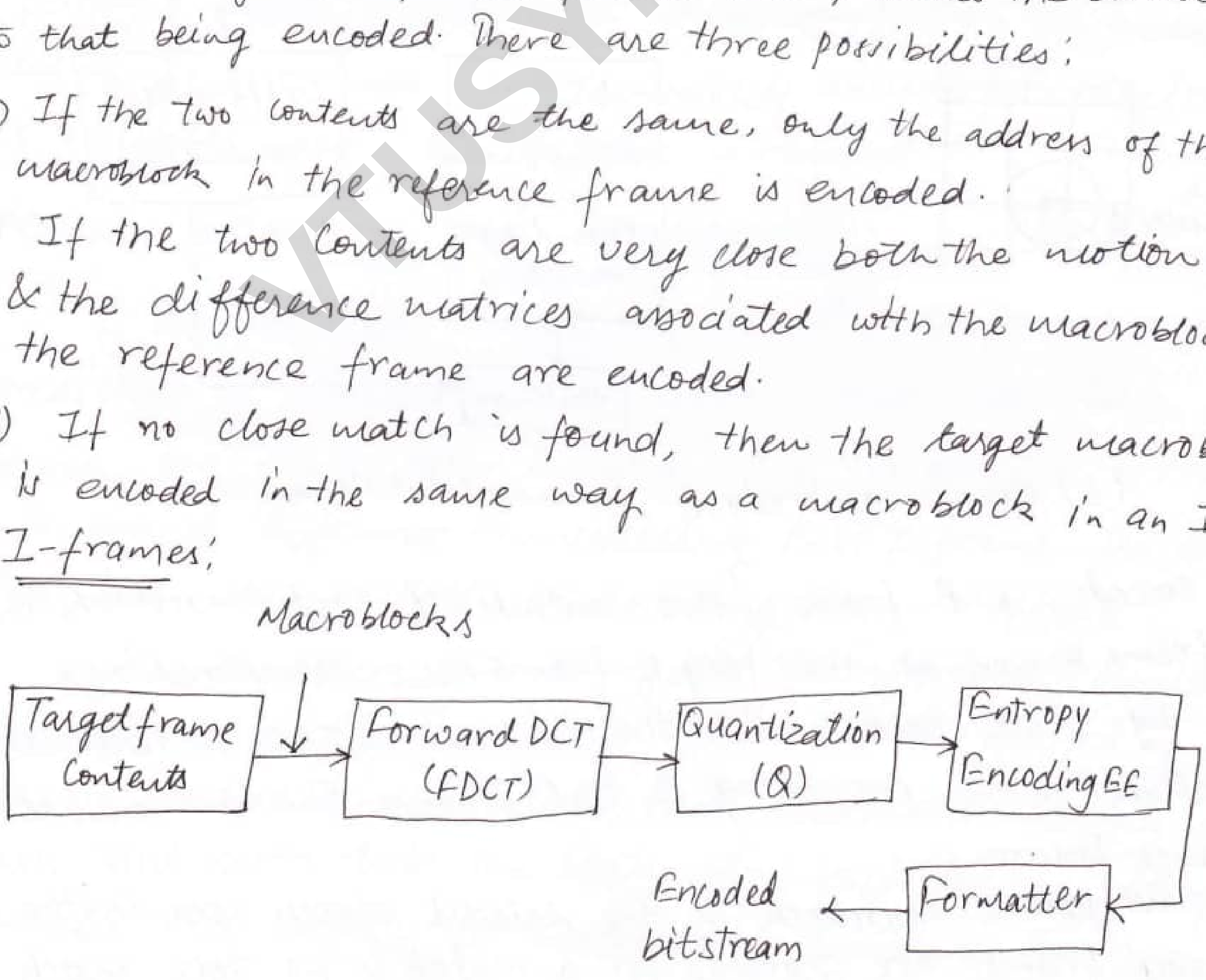
Implementation Issues;

The encoding procedure used for the macroblock that make up an I-frame is the same as that used in the JPEG standard to encode each 8×8 block of pixels. The procedure involv each macroblock being encoded using the three steps: forward DC quantization & entropy encoding.

In case of P-frames, the encoding of each macroblock is dependent on the output of the motion estimation unit depen on the contents of the macroblock being encoded & the contents of macroblock of the reference frame that produces the closest mat to that being encoded. There are three possibilities;

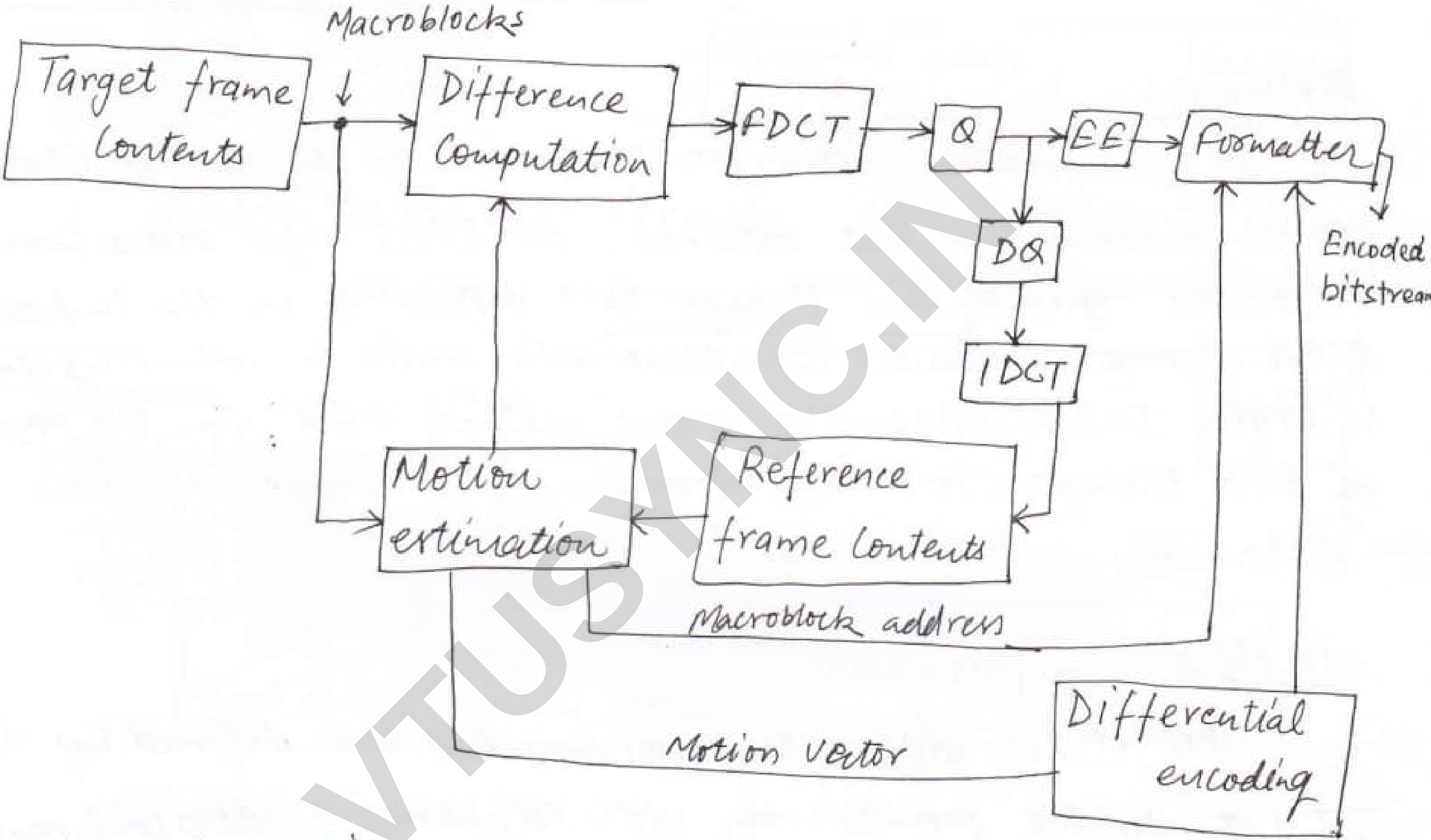
1) If the two contents are the same, only the address of the macroblock in the reference frame is encoded.
2) If the two contents are very close both the motion vecto & the difference matrices associated with the macroblock in the reference frame are encoded.
3) If no close match is found, then the target macroblock is encoded in the same way as a macroblock in an I-fra

I-frames;



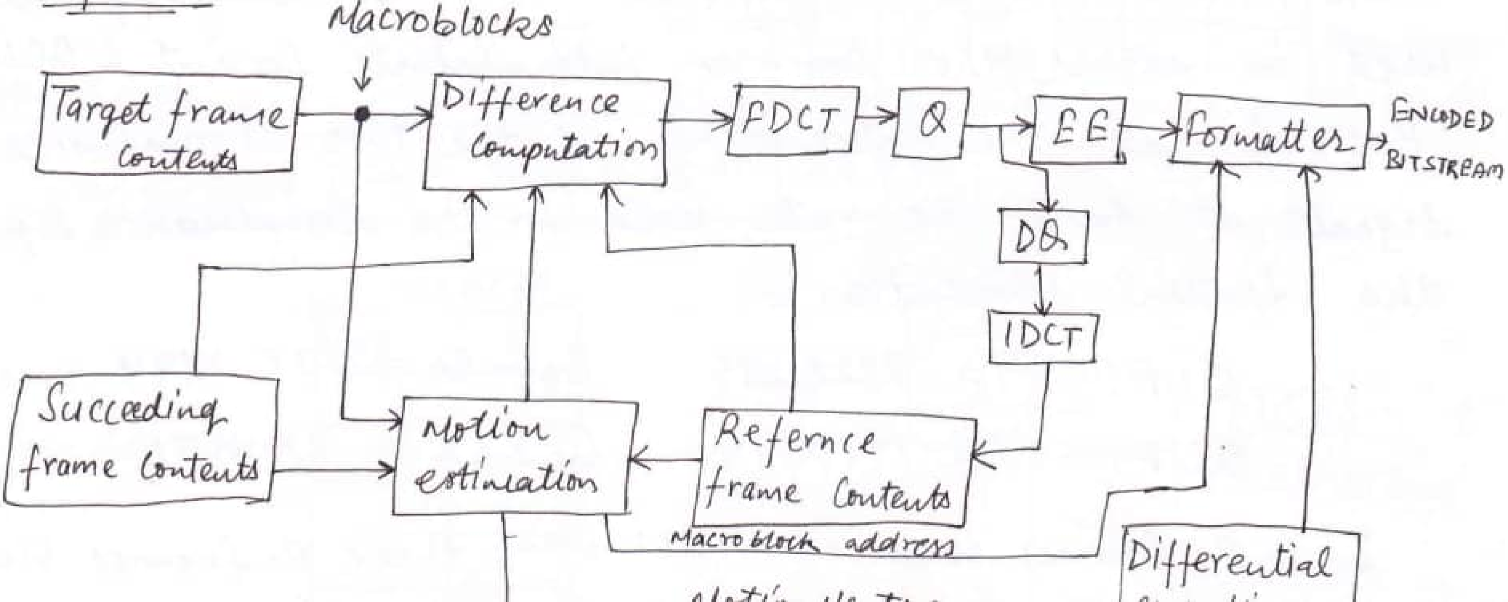
...e decoding of the received bitstream is simpler than the encoding operation since the time - consuming motion estimation processing is not required. As the encoded bits is received & decoded, each new frame is assembled a macroblock at a time. Decoding the macroblocks of an I-frame is the same as decoding the blocks in a JPEG encoded image.

P-frames:



DQ - dequantizer IDCT- inverse DCT.

B-frames:



Type	Address	Quantization value	Motion Vector	Blocks Present	B1	B2	B3	....	B6	...

DC	Skip value	Skip value	....	End of block

Performance:

The compression ratio for I-frames & hence all intracoded frames is similar to that obtained with JPEG & for video frames is in the region of 10:1 through 20:1 depending on the complexity of the frame contents. The compression ratio of both P & B-frames is higher & depends on the search method used. In the region of 20:1 through 30:1 for P-frames & 30:1 through 50:1 for B-frames.

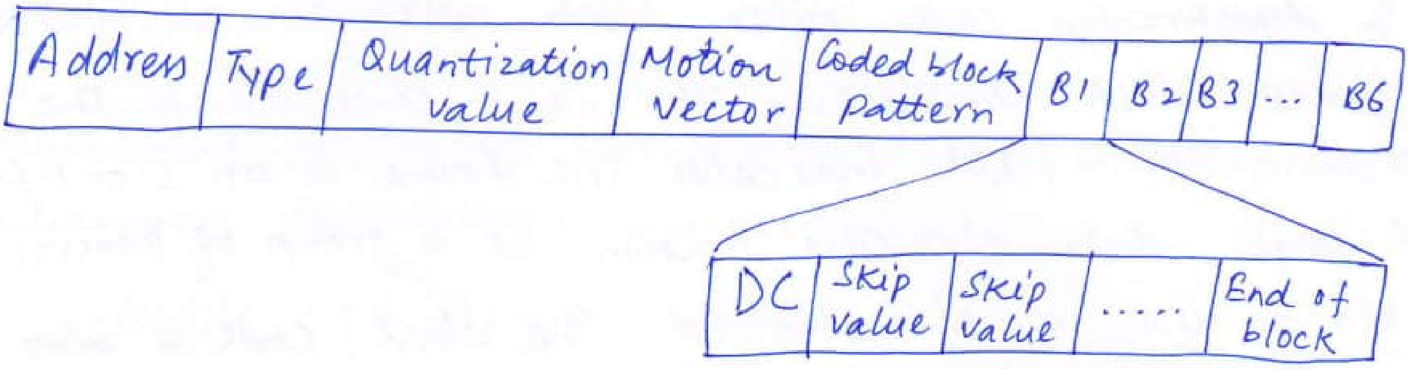
H.261:

The H.261 video compression has been defined by the ITU-T for the provision of video telephony & video conferencing services over ISDN. The standard is also known as PX64 where p can be 1 through 30. The digitization format used in either the common intermediate format & QCIF. Both formats use subsampling of the two chrominance signals at half the rate used for the luminance signal; the spatial resolution is:

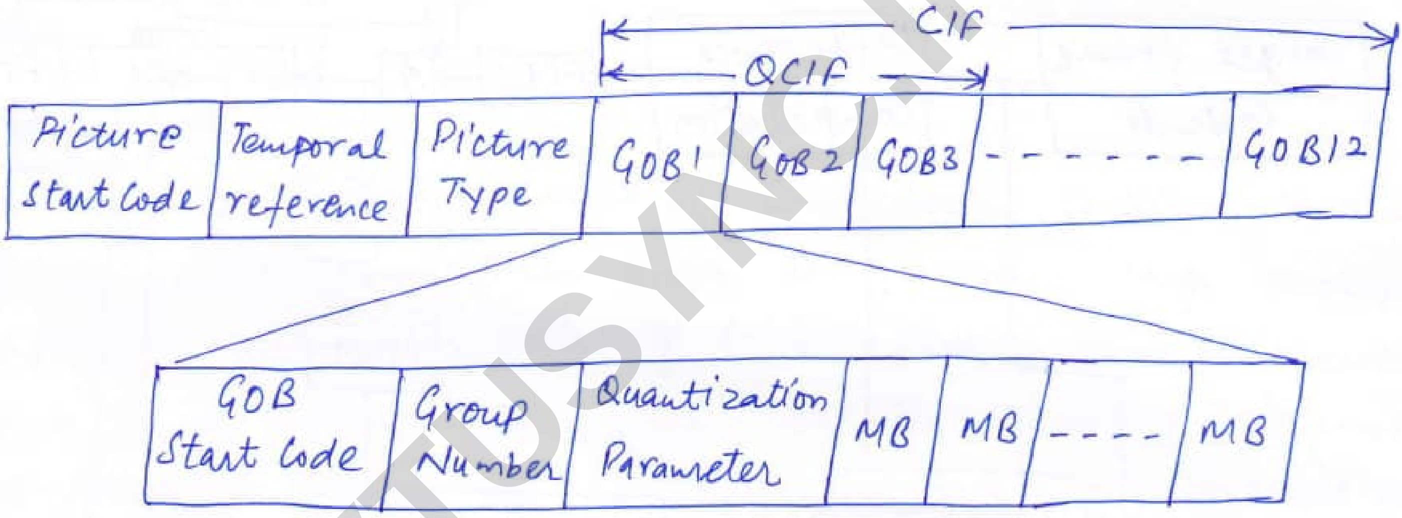
CIF: $Y = 352 \times 288$ $C_b = C_r = 176 \times 144$

QCIF: $Y = 176 \times 144$ $C_b = C_r = 88 \times 72$.

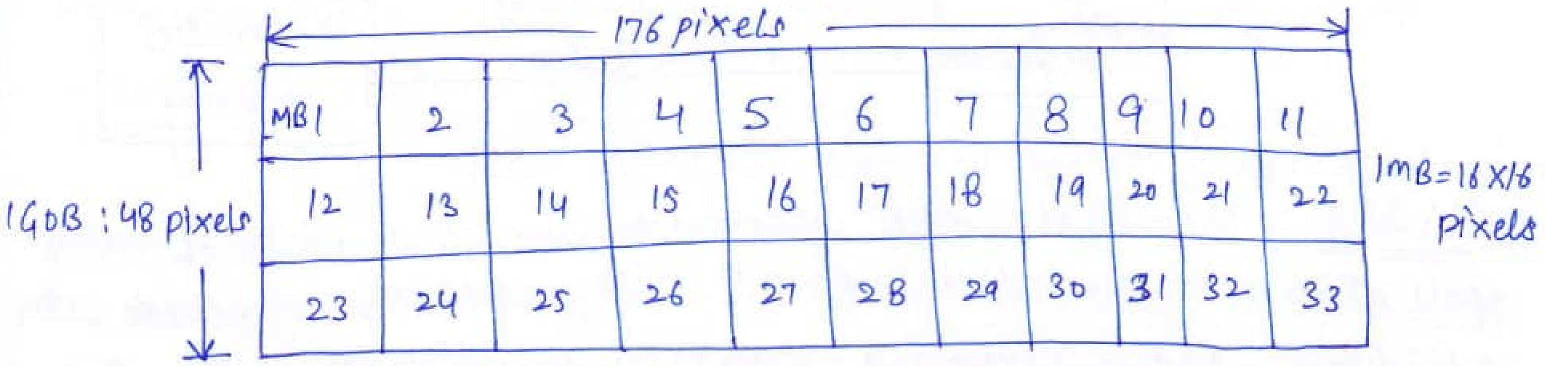
I & P-frames used in H.261 with three P-frames b/w each pair of I frames pixels blocks that make up each macroblock in both I & P-frames - 4 blocks for Y & one each for C_b & C_r is carried out using the diff. procedures.

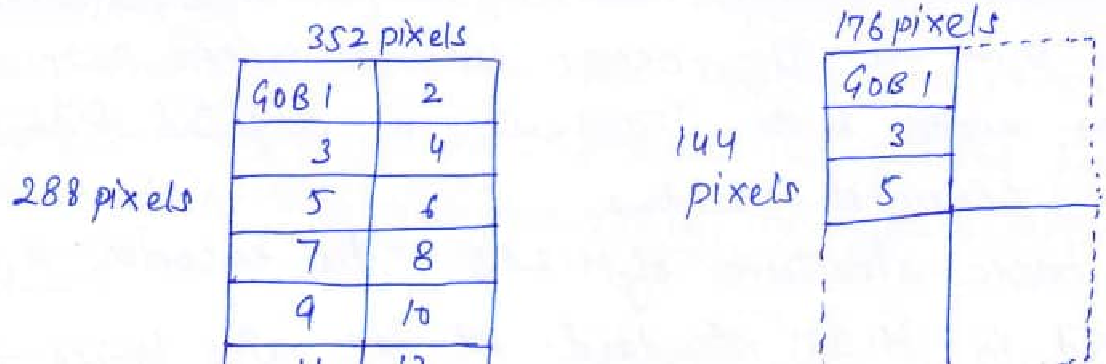


(a) Macroblock format



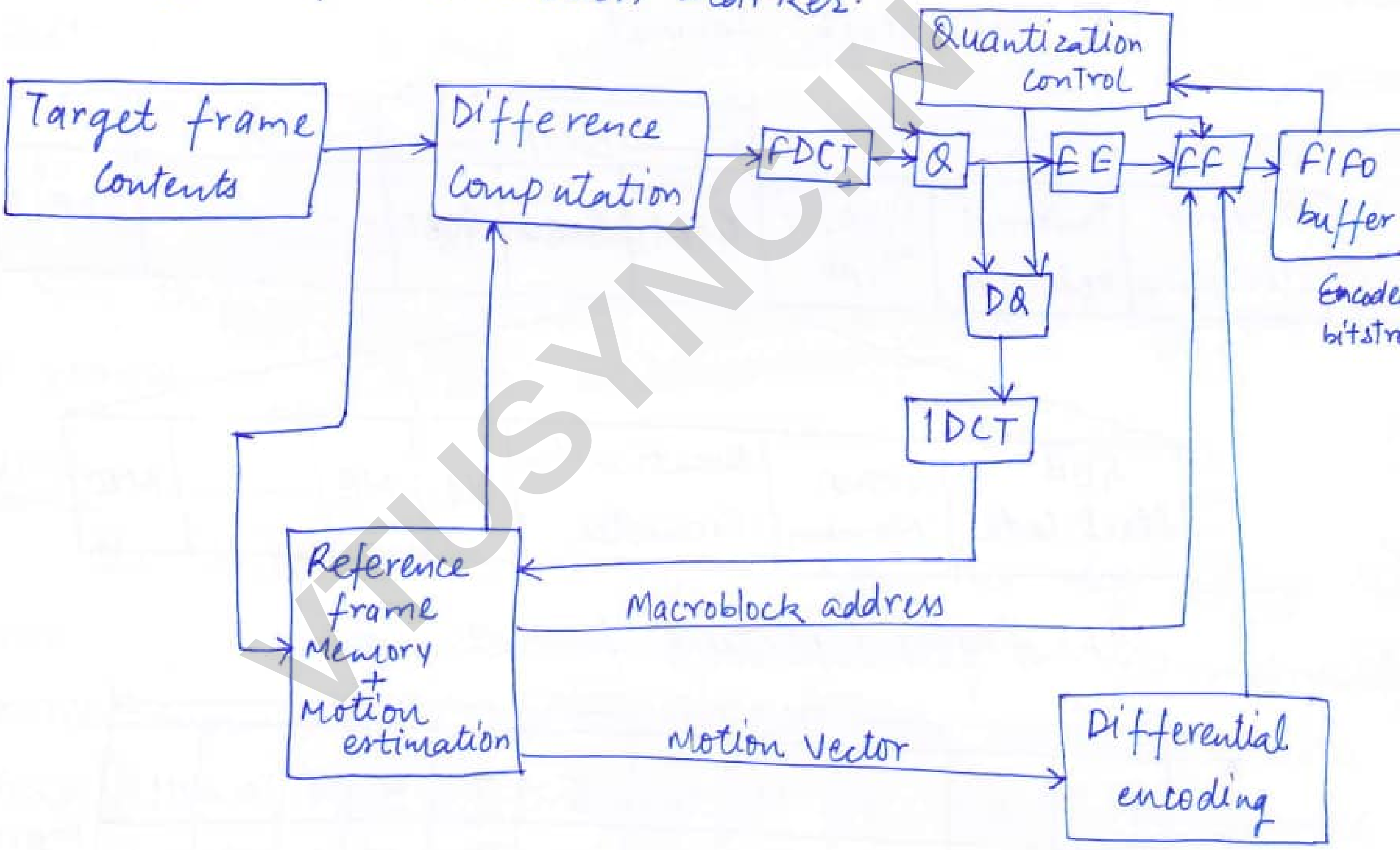
(b) frame / picture format.





(c) GOB structure

The start of each new video frame/picture is indicated b picture start code. This is followed by a temporal referen field which is a time stamp to enable the decoder to synchronize each video block with an associated audio block containing the same time stamp. The picture type field indicates the frame is an I or P-fram A larger data structure known as a group of (macro) blocks (GOB) is also defined. The start code is also known as resynchronization marker.



H.263: The H.263 video compression use in a range of video applications over wireless & PSTN. The applications include vide telephony, videoconferencing, security surveillance, interactive games playing & so on. The access circuit to the PSTN operat in an analog mode & to transmit a digital signal over these circuits, requires a modem.

The basic structure of H.263 video encoder is based on that used in H.261 standard. At bit rates lower tha

64Kbps. The high quantization threshold leads to blocking artifacts which are caused by macroblocks encoded using high thresholds differing from those quantized using lower threshold

Digitization formats:

The two mandatory formats are the QCIF & the sub-QCIF. Subsampling of the two chrominance signals is used, the two alternative spatial resolutions are:

QCIF: $Y = 176 \times 144$ $\quad C_b = C_r = 88 \times 72$

S-QCIF: $Y = 128 \times 96$ $\quad C_b = C_r = 64 \times 68$

Progressive scanning is used with a frame refresh rate of either 15 or 7.5 fps.

Frame types:

To obtain higher levels of compression, H.263 standard use I-, P & B-frames. In order to use as high a frame rate as possible neighbouring pairs of P- & B-frames can be encoded as a single entity. The resulting encoded frame is known as PB-frame & because of much reduced encoding overheads that are required, its use enables a higher frame rate to be used with a given transmission channel.

Unrestricted motion Pictures:

The motion vectors associated with predicted macroblocks are normally restricted to a defined area in the reference frame around the location in the target frame of the macroblock being encoded. A small portion of potential code match macroblock fall outside of the frame boundary, then the target macroblock is automatically encoded as for an I-frame. This occurs even through the portion of the macroblock within the frame area is a close match.

To overcome this limitation, inunrestricted motion vector mode, for those pixels of a potential close-match macro-block that fall outside of the frame boundary, the edge pixels themselves are used.

Error Resilience:

This type of network is relatively high probability that transmission errors will be present in the bitstream received by the decoder. Errors are characterized by periods when a string of error-free frames is received followed by a short burst of errors corrupt a string of macroblocks within a frame. The related group of (macro) blocks (GOB) contains one or more macroblocks that are in error.

When an error in GOB is detected, the decoder skips the remaining ~~maskes~~ macroblocks in the affected GOB & searches for the unique resynchronization marker at the head of the next GOB. It then decodes from the start of GOB. An error concealment scheme incorporates the decoder.

Error Tracking:

With real-time applications such as video telephony, a two way communication channel is required for the exchange of the compressed audio & video information generated by the codec in each terminal. There is always a return channel from the receiving terminal back to sending terminal & this is used in all three schemes by the decoder to inform the related encoder that an error in a GOB has been detected. Errors are detected in a no. of ways including:

⇒ one or more out of range motion vectors.
⇒ one or more invalid variable - length codewords.
⇒ one or more out-of-range DCT Coefficients.
⇒ an excessive no. of coefficients within a macroblock.

Independent Segment decoding:

The aim of this scheme is not to overcome errors that occur within a GOB but rather to prevent these errors from affecting neighbouring GOBs in succeeding frames. To achieve this each GOB is Treated as a ~~sequence~~ separate subvideo which is independent of other GOBs in the frame.

When an error in a GOB occurs the same GOB in each successive frame is affected & neighbousing GOBs are not affected.

Reference Picture Selection:

Error tracking scheme in as much as it endeavors to stop errors propagating by the decoder returning acknowledgement messages when an error in a GOB is detected. It Can be operated in two different modes.

The operation i.e NAK operation mode, in this mode, only GOBs in error are signaled by the decoder returning a NAK message. During the encoding of intercoded frames a copy of preceding frame is retained by encoder. The encoder can select any of these previously decoded frames as the reference. When a NAK is received, the encoder selects GOB 3 of frame 1 as the reference to encode GOB 3 of the nxt frame

The alternative mode of operation is known as ACK mode. In this all frames received without errors are

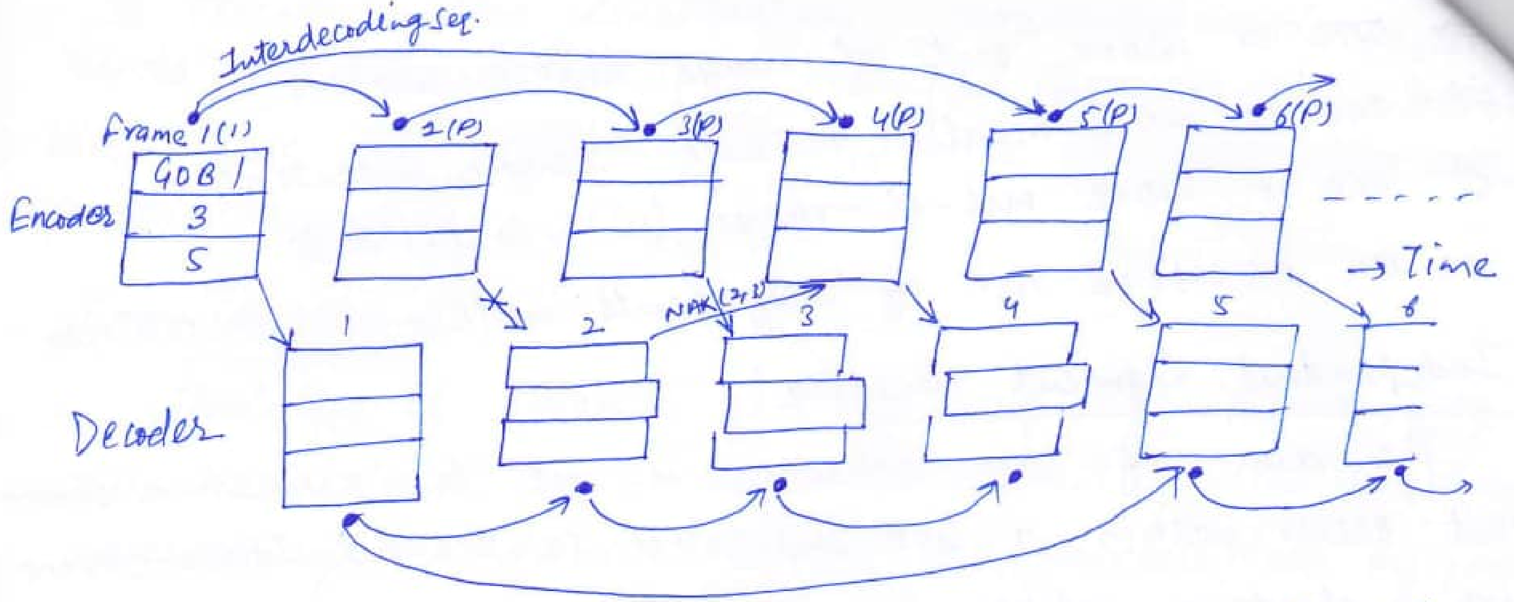


fig. Referance picture selection with independent segment decoding: NAK mode.

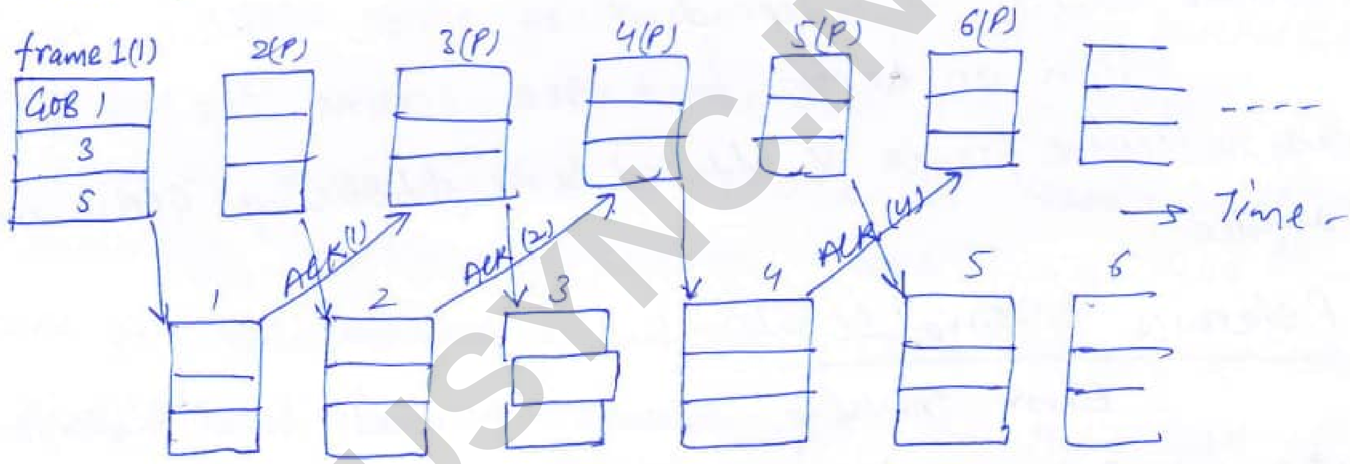


fig: Reference picture selection with independent Segment decoding: ~~NA~~ACK mode.

acknowledged by the decoder returning an ACK message. Only frames that have been acknowledged are used as reference frames. The ACK mode performs best when the round trip delay of the Communication channel is short & ideally less than the time the encoder takes to encode each frame.

MPEG: Motion Pictures Expert Group:

MPEG is ~~a~~ related to a range of multimedia applications that involve the use of video with sound. The outcome is a set of three standards which relate to either the recording or the transmission of integrated audio & video streams. Each is targeted at a particular application domain. The three standards, which use different video resolution are:

⇒ **MPEG-I**: This is defined in a series of documents which are all subsets of ISO Recommendation 11172. The video resolution is based on the source intermediate digitization format (SIF) with a resolution of up to 352×288 pixels. The standard is intended for the storage of VHS-quality audio & video on CD-ROM at bit rates up of 1.5Mbps. Normally, higher bit rates of multiples of this are more common in order to provide faster access to the stored material.

⇒ **MPEG-2**: This is defined in a series of documents which are all subsets of ISO Recommendation 13818. It is intended for the recording & transmission of studio-quality audio & video. The standard covers four levels of video resolution:

→ Low: based on the SIF digitization format with a resolution of upto 352×288 pixels. It is Compatible with the MPEG-I standard & produces VHS-quality video. The audio is of CD quality & target bit rate upto 4Mbps

→ Main: based on the 4:2:0 digitization format with a resolution of upto 720×576 pixels. This produces studio-quality digital video & the audio allows for multiple CD-quality audio channels. The target bit rate is upto 15Mbps or 20Mbps with the 4:2:2 digitization format.

→ High 1440: based on the 4:2:0 digitization format with a resolution of 1440 X1152 pixels. It is intended for HDTV at bit rates up to 60 Mbps or 80 Mbps with the 4:2:2 format

→ High: based on the 4:2:0 digitization format with a resolution of 1920X1152 pixels. It is intended for wide-screen HDTV at a bit rate of upto 80 Mbps or 100Mbps with the 4:2:2 format

MPEG-4:

Initially, this standard was concerned with a similar range of applications to those of H.263, each running over very low bit rate channels ranging from 4.8 to 64 Kbps. Its scope was expanded to embrace a wide range of interactive multimedia applications over the Internet & the various types of entertainment networks.

Recommended Questions

1. Explain the process of MPEG audio encoding?[10]
2. Explain the MPEG data stream?[06]
3. Explain video coding extension w.r.t MPEG-4?[06]
4. State the imp components of MPEG-4 terminal with DMIF architecture?[07]
5. What is fractal compression?[04]
6. Explain the process of scaling compression video w.r.t. MPEG-2?[10]
7. Explain the key differences b/w MPEG-2 , MPEG-4, MPEG?[08]
8. Explain the process of scaling w.r.t MPEG?[05]
9. Explain the sectional view of an optical disc along the data track? Give an Ex for an optical disc ?[05]
10. Define video disc and WORM along with their properties?[06]
11. Write the standards defined by MPEG 1[05]
12. Write the standards defined by MPEG 2[05]
13. Write the standards defined by MPEG 4[05]
14. What are I frames, P frames, B frames? [05]
15. What is SIF? [05]
16. What is CIF? [05]
17. Define brightness, hue, saturation. [05]
18. What is digital video editing? [05]
19. What is digital video post production? [05]
20. Compare vector graphics and bitmapped images[08]
21. Write short notes on JBIG.
22. Define pitch, loudness.[04]
23. Write the types of voice track excitation parameters.[04]
24. Define frequency masking.[05]
25. Define temporal masking.[04]
26. What is LPC?[04]
27. What is GIF?[04]
28. Write the standard defined by TIFF.[06]
29. Explain in detail the digitalization of video signals. Also Explain in detail the various digitisation formats of video signal.[10]
30. Explain in detail MPEG 4 coding principles.[10]
31. Explain in detail the principles of video compression.[10]
32. Explain in detail digitization of sound.[10]
33. Explain in detail various types of audio compression techniques.[10]
34. Explain in detail a) LPC encoder and Decoder b) Perceptual compression[10]
35. Explain in detail JPEG encoder and decoder[10]
36. Explain in detail various standards associated with image compression.[06]
37. Write short notes on a) MIDI b) JBIG[10]